

Account Age VS Tone of Voice



Introduction

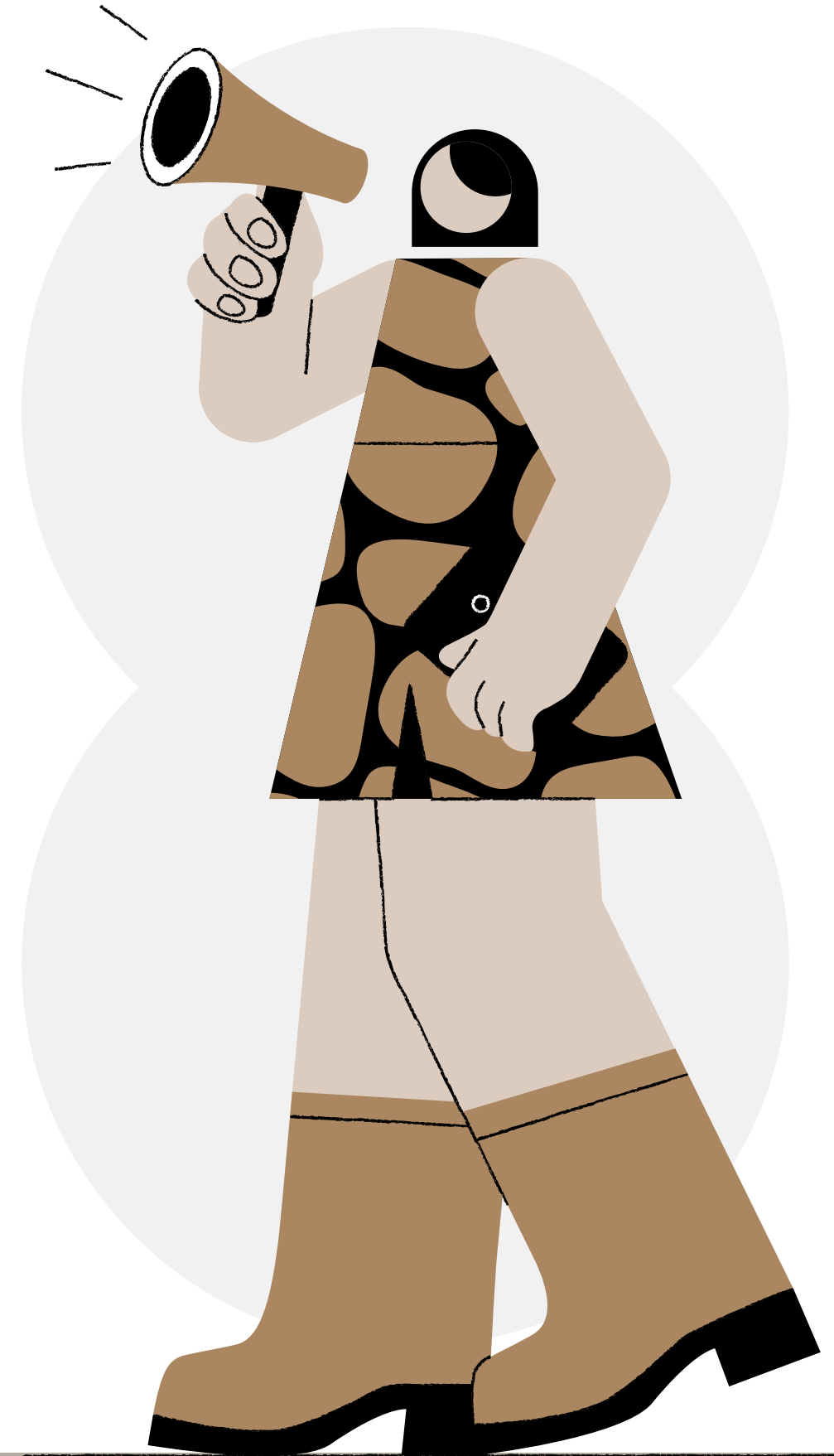
This project analyzes Reddit comments to understand the factors influencing online toxicity.

Originally, we aimed to explore how account age affects toxicity.

However, after deeper analysis, we shifted toward understanding how subreddit context and comment text influence toxic behavior.

Our goal:

Build a machine learning model that predicts whether a comment is toxic using NLP and metadata features.





Research Question

How does the age of a user's account correlate with the likelihood of them posting a toxic comment?

Do certain Topics have a significantly higher proportion of toxic comments than others?

Can we accurately predict the toxicity of a comment based on account age?

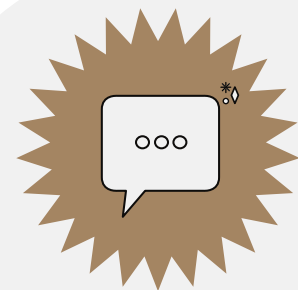
Outline



Data
Collection



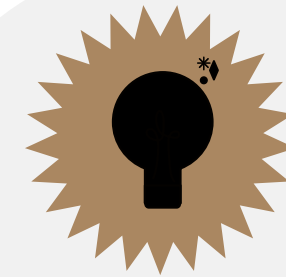
Data
Preprocessing



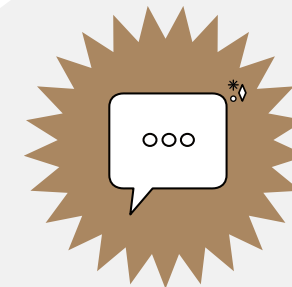
Exploratory
Data Analysis



Modelling



Evaluation
Metrics



Technical
Hurdles



Model
Improvement
&
Future Work

Data Collection

Data Collection Methods

- PRAW (Reddit API Wrapper) – used to fetch comments, authors, subreddits, and account age
 - **Python Libraries:**
 - pandas (structuring data)
 - datetime (age calculation)
 - time (managing API rate limits)
-

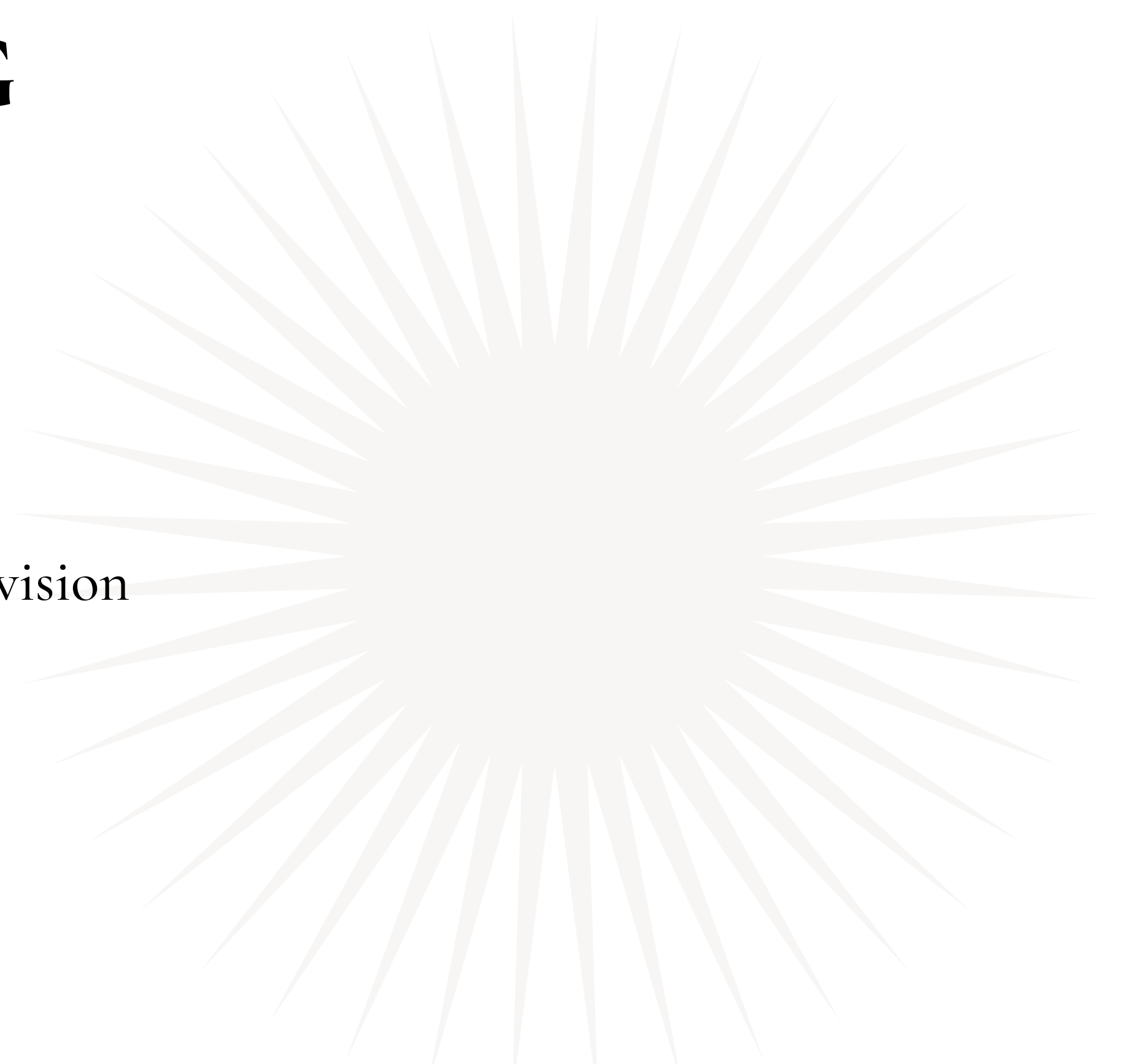
We collected 601 comments across 3 subreddits:

- r/politics
- r/gaming
- r/movies



DATA PREPROCESSING

- Handling missing and duplicate data
- Cleaning and standardizing comment text
- Generating toxicity labels using weak supervision
- Saving the processed dataset as a CSV



Exploratory Data Analysis

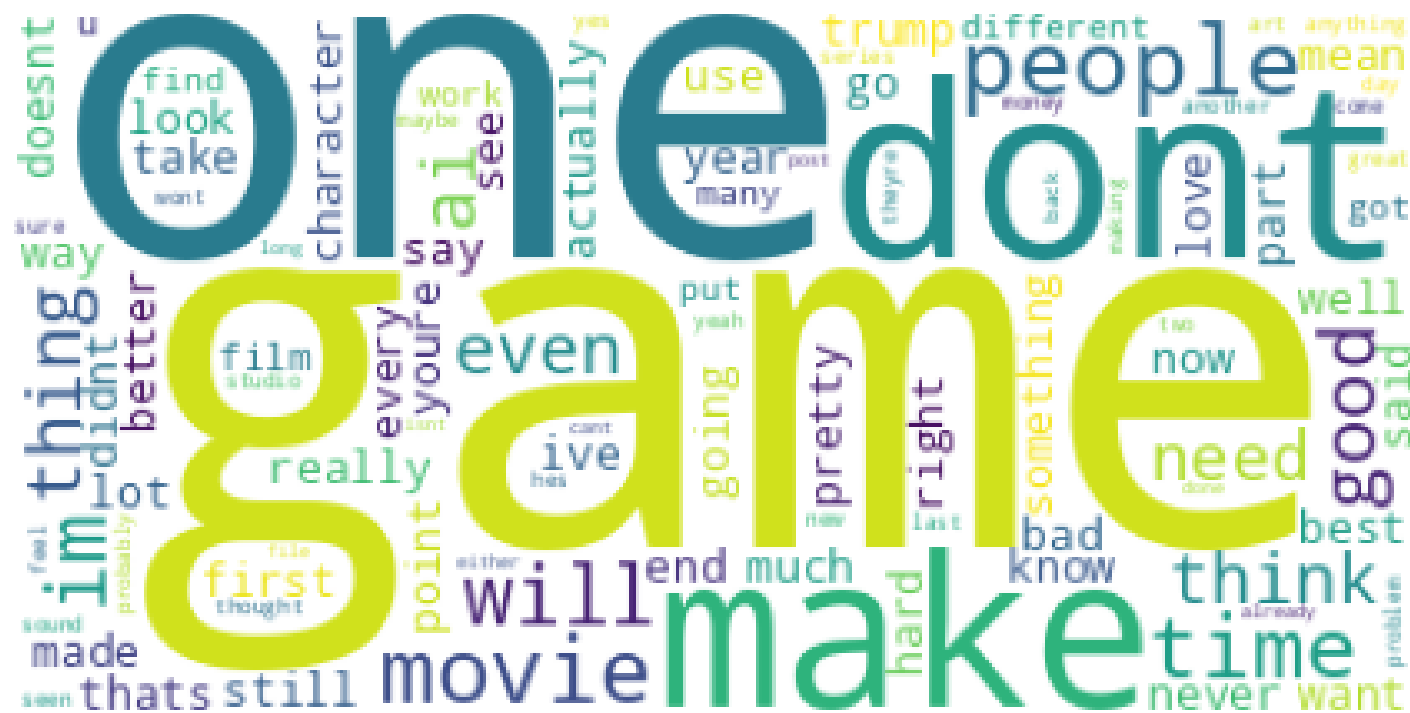
- Toxic comments = ~10%, confirming strong class imbalance
- r/politics shows highest toxicity
- Comment lengths: toxic comments tend to be shorter
- Correlation matrix: weak correlations between account age and toxicity
- Word clouds:
 - Non-toxic: neutral/common words
 - Toxic: aggressive/offensive wording

Tools Used: Matplotlib, Seaborn

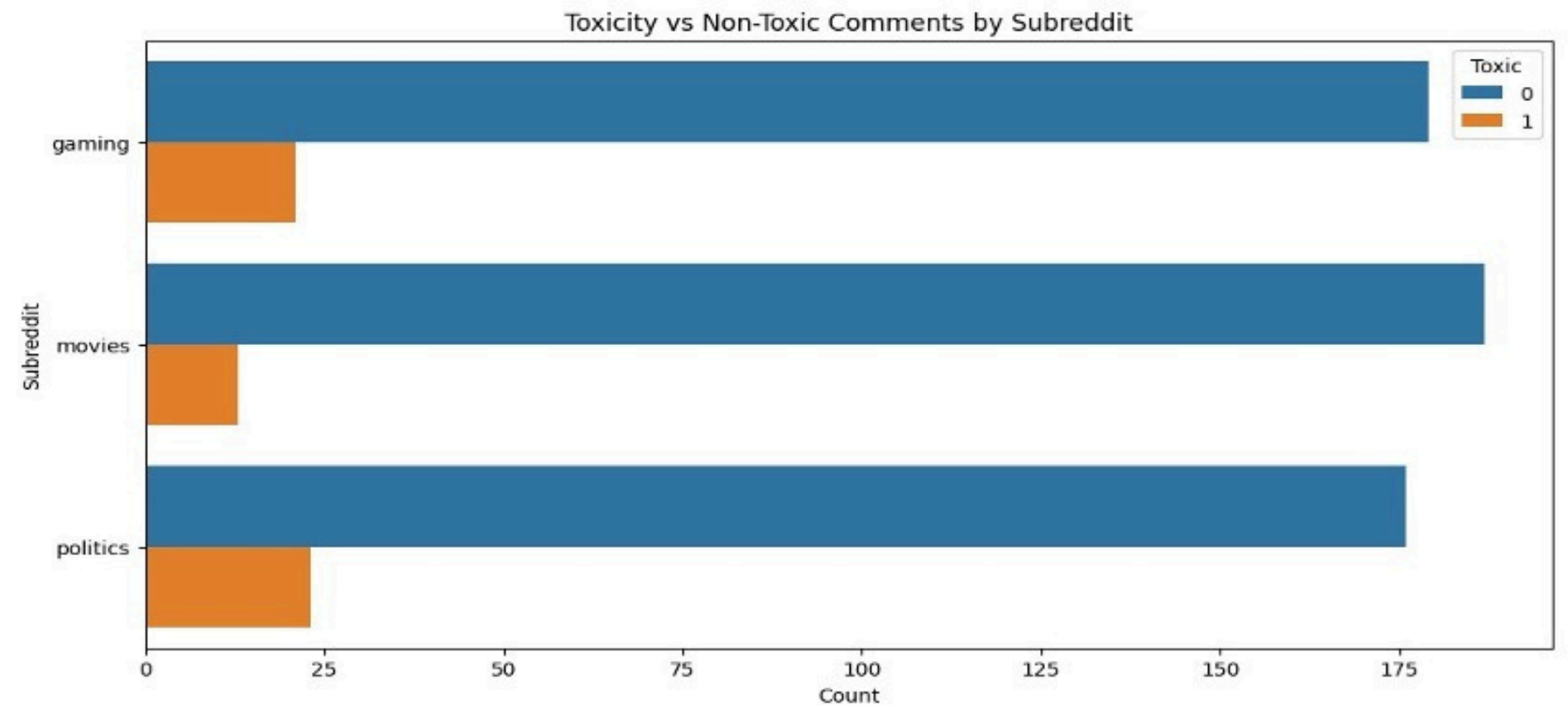
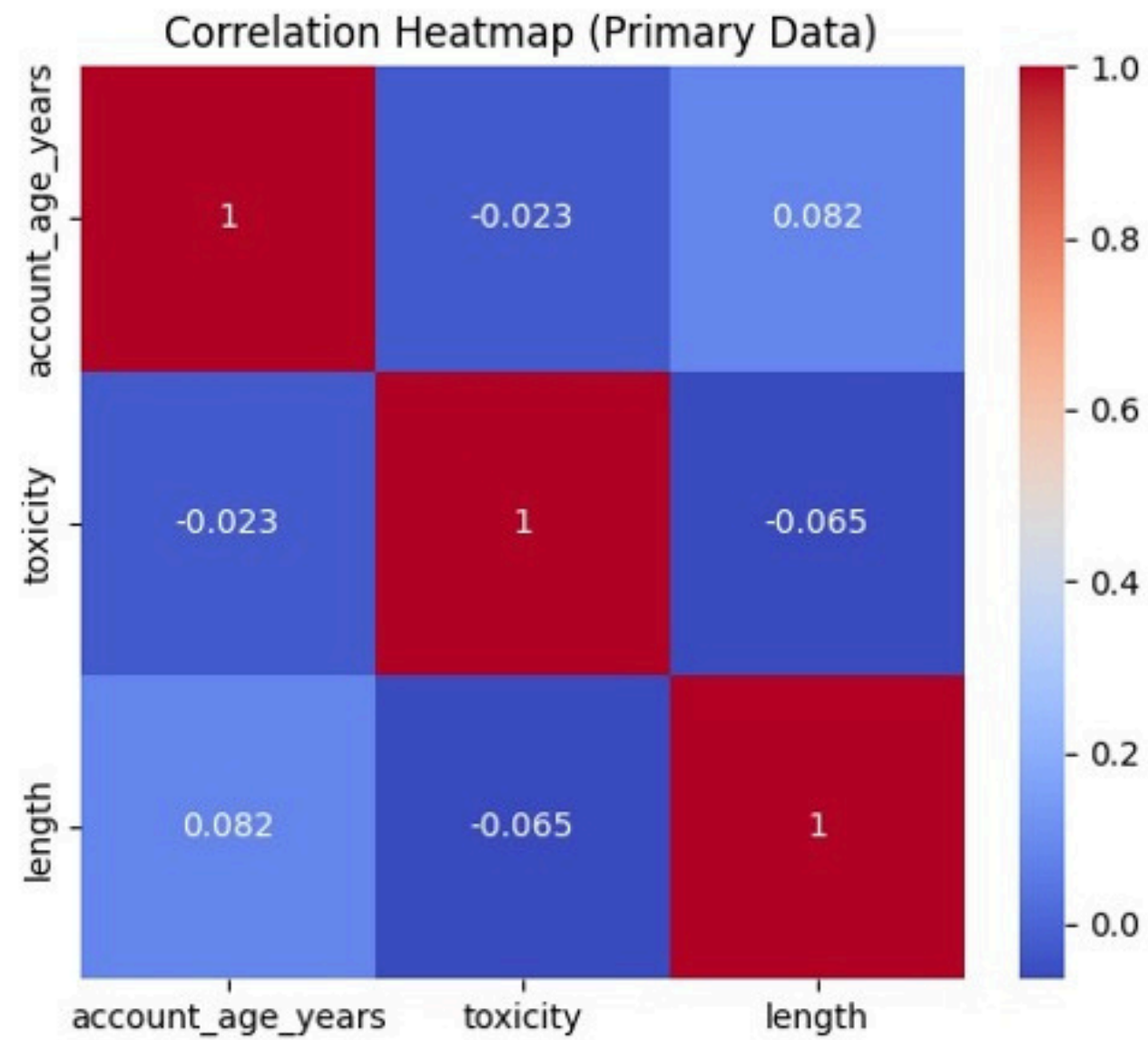
Word Cloud - Toxic Comments



Word Cloud - Non-Toxic Comments



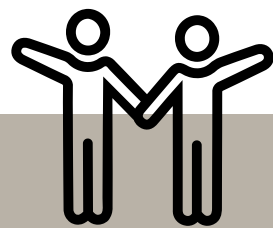
Exploratory Data Analysis



Modelling Overview

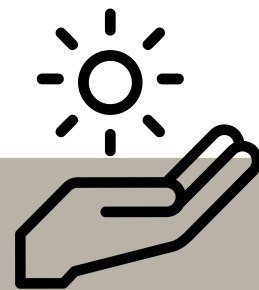
Most Baseline Model

**Logistic Regression
(account_age only)**



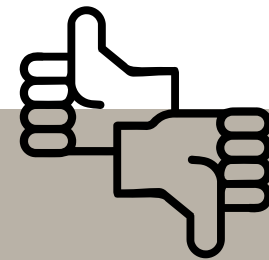
Model 2

**Logistic Regression +
TF-IDF + Subreddit +
Age**



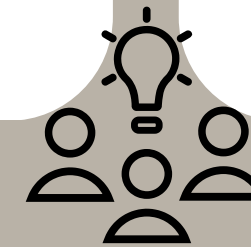
Model 3

**Linear SVM + TF-
IDF + Subreddit +
Age**



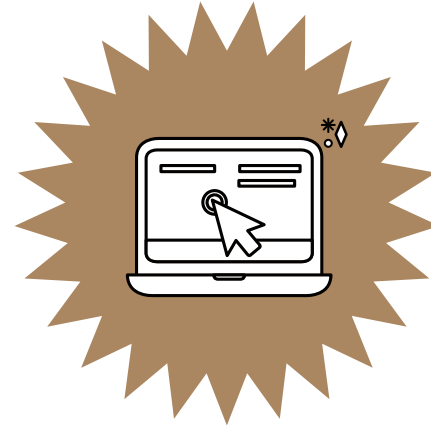
Feature Engineering:

- Comment Text → TF-IDF vectorization
- Subreddit → One-Hot Encoding
- Account age → Standardized



**Used
class_weight="balanced" due to
imbalance.**

ALGORITHMS



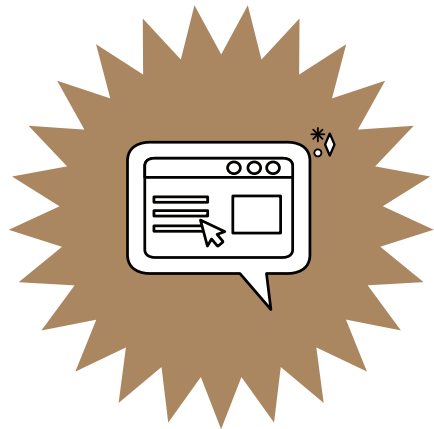
Baseline Logistic Regression

- Strong classifier for high-dimensional text.
- Maximizes margin between classes.



Logistic Regression + TF-IDF

- Works extremely well with high-dimensional sparse text.
- Lightweight, interpretable, and ideal for small datasets.



Linear SVM + TF-IDF

- Strong classifier for high-dimensional text.
- Maximizes margin between classes.

BASELINE MODEL RESULTS				

	precision	recall	f1-score	
0	0.87	0.54	0.67	
1	0.04	0.18	0.06	
accuracy				0.51

LOGISTIC REGRESSION + TF-IDF RESULTS				
	precision	recall	f1-score	
0	0.94	0.98	0.96	
1	0.67	0.36	0.47	
accuracy				0.93

LINEAR SVM + TF-IDF RESULTS				

	precision	recall	f1-score	
0	0.93	0.98	0.96	
1	0.60	0.27	0.38	
accuracy				0.92

Accuracy:

- Not reliable due to class imbalance (~90% non-toxic).

Recall (Toxic):

- Measures how many toxic comments were caught.
- LR = 0.36, SVM = 0.27, Baseline = 0.18.

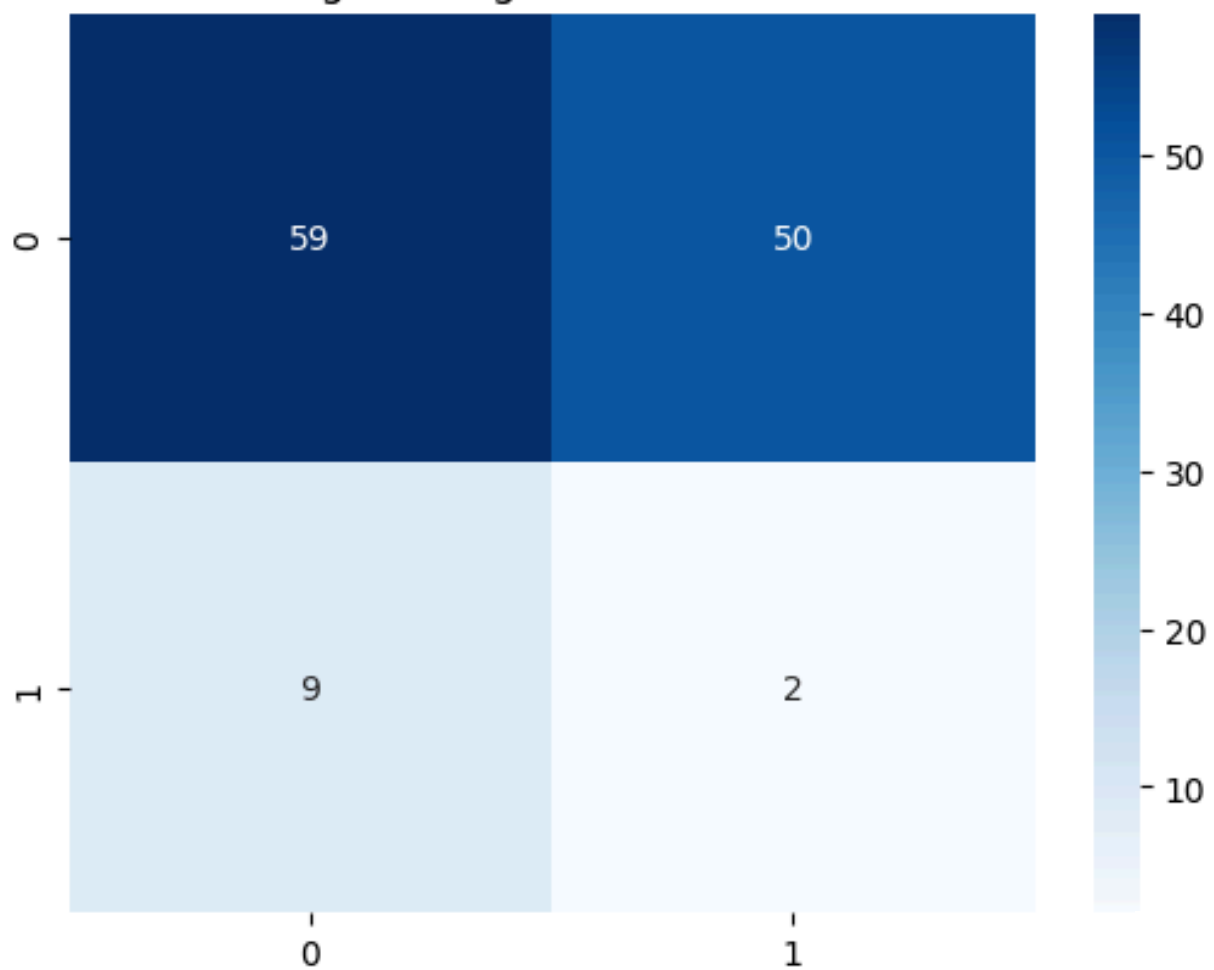
Precision (Toxic):

- Measures correctness of toxic predictions.
- LR = 0.66 (highest), SVM = 0.60, Baseline = 0.038 (very low).

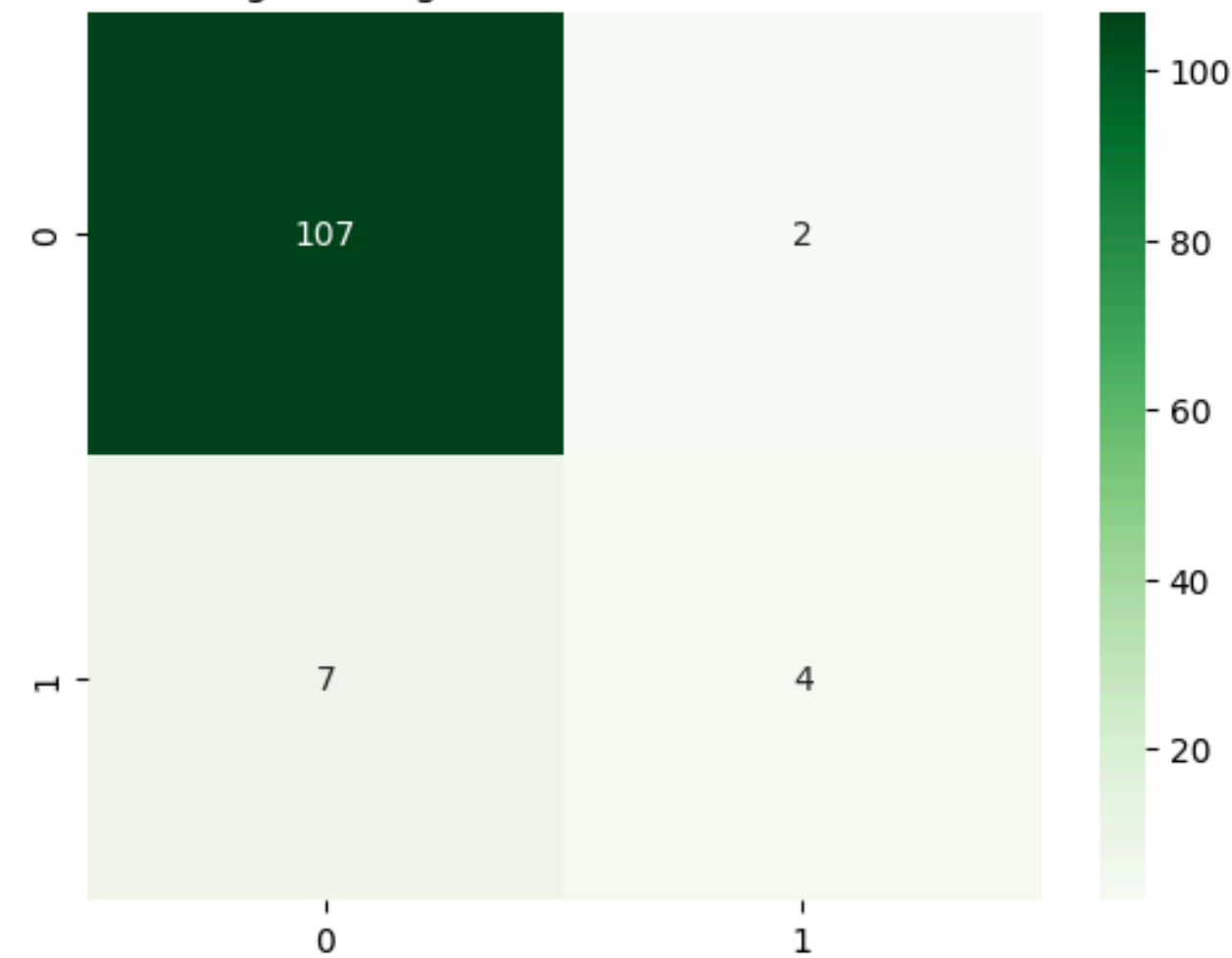
F1-Score:

- Best metric for imbalanced datasets (balances precision + recall).
- LR = 0.47 → best overall model; SVM lower; Baseline poor.

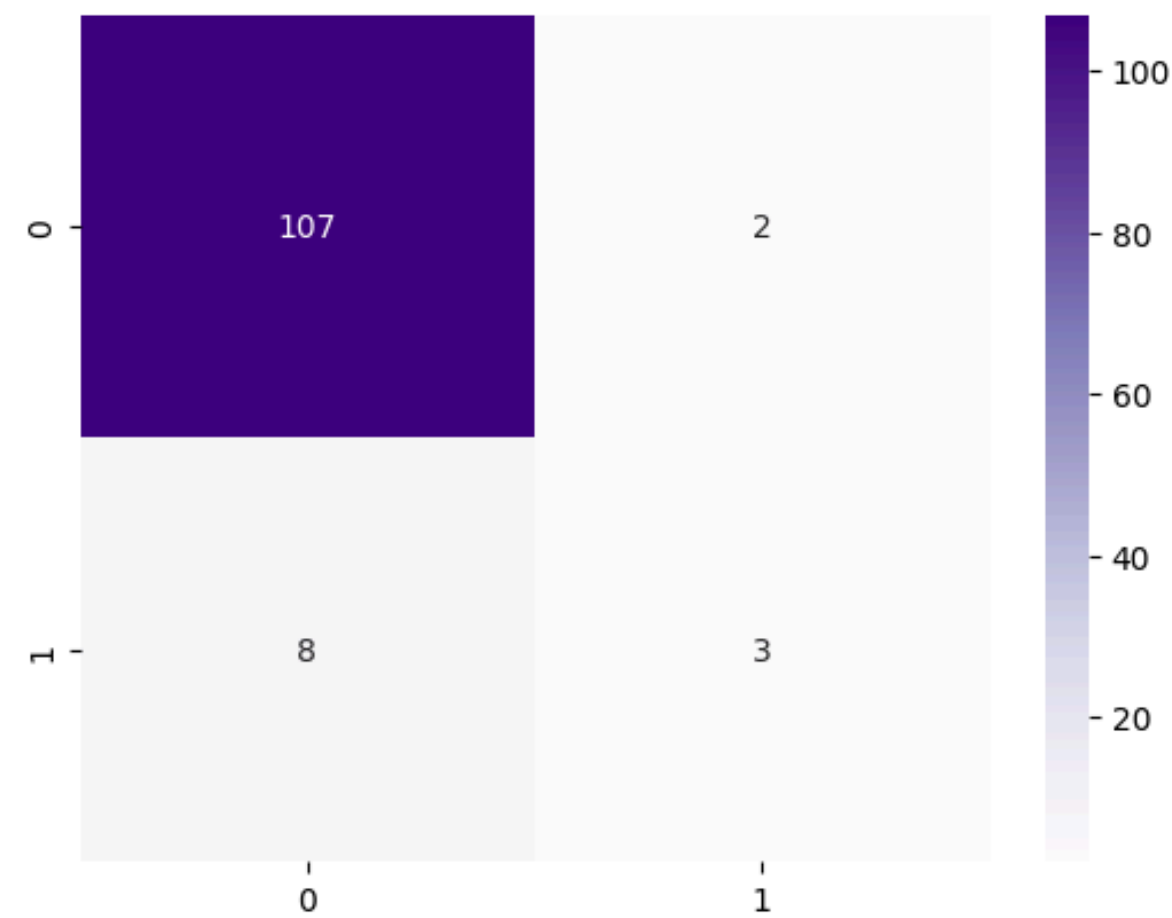
Baseline Logistic Regression Confusion Matrix



Logistic Regression Confusion Matrix



Linear SVM Confusion Matrix



Evaluation Metrics

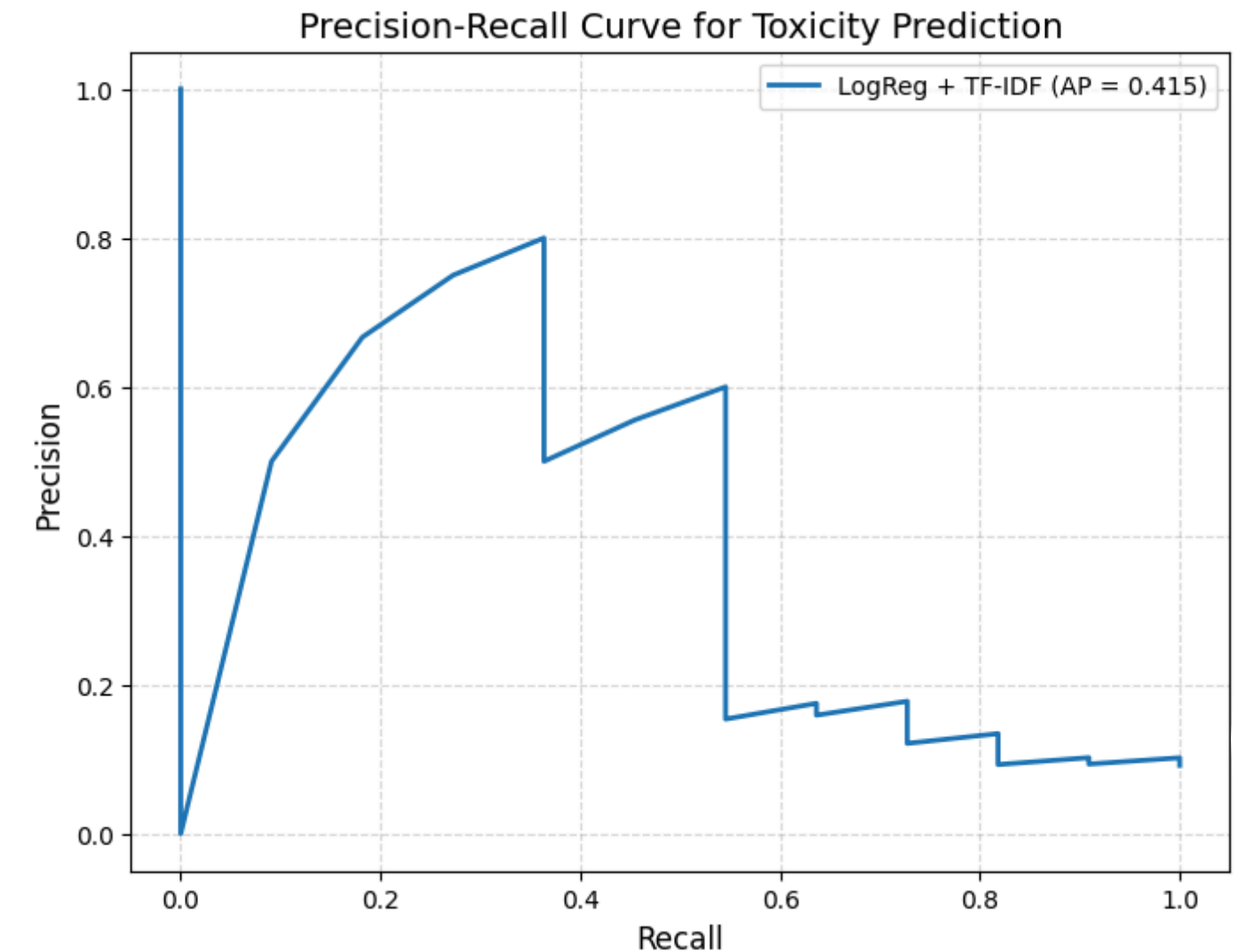
KEY FINDINGS :

Baseline Model (accountAge Only)

- $F_1 = 0.06 \rightarrow$ Almost no ability to detect toxic comments
- Shows account age is NOT a meaningful predictor
- Essential benchmark to justify advanced models

Logistic Regression + TF-IDF (Best)

- $F_1 = 0.47$ (highest)
 - $AP = 0.415 \rightarrow$ Best performance under imbalance
 - Strong pattern detection from text & subreddit features
-
- Precision–Recall Curve / AP:
 - Most appropriate metric for minority class.
 - LR $AP = 0.415$, confirming strong performance.



Results Visualization

Include :

- * Confusion Matrix (LR model)
- * Precision-Recall Curve (AP = 0.415)
- * Model comparison table

Talking points:

- * Strong differentiation between classes
- * Good performance under imbalance
- * Best ability to detect toxic content

TECHNICAL HURDLES

- Reddit API rate limits
- Small datasets
- Class imbalance (90% non-toxic)
- Text noise
- Weak labels from retrained model



HOW TO OVERCOME TECHNICAL HURDLES?



Reddit API rate limits

implemented back-off timing to avoid request failures



Small datasets

chose lightweight models to prevent overfitting



Class imbalance (90% non-toxic)

applied class weighting to improve fairness



Text noise

performed thorough cleaning for better model input



Weak labels from retrained model

ensured label consistency via manual checks



How to Improve the Model

potential Model Improvement:

- Collect larger dataset for stronger generalization

- Fine-tune threshold to improve recall

- Train topic-specific models (per subreddit)

Conclusion

- Original question refined to focus on subreddit influence
- Built three models for toxicity prediction
- TF-IDF Logistic Regression achieved best performance
- Subreddit + text far more important than account age
- Successfully built a reliable toxicity classifier



Thank You

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